

Borrowing From the Worm

What the largest oxygen-carrying molecules on Earth can teach the people building the next BHOC

Human hemoglobin is a small protein. Four globin chains, four hemes, about 64,000 daltons. It works because it is packed by the hundreds of millions into a red cell, where a membrane, a battery of enzymes, and a bath of the small molecule 2,3-BPG keep it in working order. Take it out of the cell and the whole arrangement falls apart. The molecule splits into halves, leaks through the kidney filter, grabs nitric oxide in the vessel wall, and rusts.

Almost every effort to build a hemoglobin-based oxygen carrier has been an effort to fix those four problems by force: cross-link the halves, polymerize the pieces, wrap them in polymer, or shrink them into a vesicle. Bovine hemoglobin (the BHOC route) inherits the same logic. It is a clever way to repair a small protein that was never meant to live alone.

There is a different starting point. Some animals never put their hemoglobin in cells at all. They dissolve it, in giant assemblies, directly in their blood. Those molecules have been doing what the engineers want for hundreds of millions of years. This post is about what they look like, why they work, and what a careful reader should and should not conclude from them.

A molecule the size of a virus

The common earthworm, *Lumbricus terrestris*, carries an oxygen protein called erythrocrucorin. It is not a hemoglobin in the sense we usually mean, although it is a relative. It has a molecular mass of roughly 3.6 million daltons, around fifty-five times heavier than human hemoglobin. It is built as two stacked hexagonal rings, with something on the order of 144 globin chains arranged around a scaffold of linker chains. Under an electron microscope the particle looks like a small, flat, six-fold flower about thirty nanometers across.

That size is the point. A molecule of that mass cannot pass through the kidney filter. It cannot slip between the cells lining a vessel and wander into the tissue, where it would scavenge nitric oxide and squeeze the surrounding vessels shut. It does not dissociate into dimers, because it is held together by many contacts, and in the earthworm it is stabilized by calcium. It carries dozens of oxygen molecules per particle. It is a molecule that behaves the way a small red cell would, minus the cell.

The worm has other advantages. Its erythrocrucorin is cooperative, meaning it binds oxygen with the sigmoidal curve that allows efficient loading and unloading. Studies on the purified protein show it holds up to oxidation more gracefully than many engineered hemoglobins, and it does not require a cell to keep it reduced. I would treat that claim with normal care: the data come from a small number of laboratories and mostly from preclinical settings. But the direction of the evidence is consistent.

The lugworm that walked into an operating room

The marine lugworm, *Arenicola marina*, carries a similar giant hemoglobin. A French company, Hemarina, has developed a product from it, often called M101 or HEMO2life. The company reports that the molecule carries far more oxygen per particle than human hemoglobin and, notably, that it has an enzyme-like ability to neutralize reactive oxygen species. The stated rationale is protection of organs kept outside the body before use.

In 2016 the product was used in France in a first-in-human case as a preservation additive for a donated kidney. Since then it has been tested in a small number of further studies. I am deliberately keeping the description at that level. The published human experience is small, the design is not a randomized comparison of survival, and much of what is said about it comes from the developer. What deserves attention is not the marketing. It is that a giant, cell-free, non-human oxygen carrier has been put through a regulated human procedure at all.

Two things follow. First, the safety signal for the worm molecule in people, so far, is clean enough that studies have continued. Second, and more important for the BHOC conversation, it demonstrates the principle that a very large hemoglobin can be given to a human without the vasoconstriction that defined earlier products. That is a proof of mechanism, not of benefit.

The tubeworm that breathes poison

The strangest example belongs to *Riftia pachyptila*, the giant tubeworm of deep-sea hydrothermal vents. It lives beside water rich in hydrogen sulfide, a compound that shuts down mitochondrial respiration in almost every other animal. It has no mouth and no gut. Its tissues contain bacteria that oxidize sulfide for energy, and the worm feeds the bacteria by delivering both oxygen and sulfide to them at the same time.

Doing that in a single bloodstream is a chemical trick. Sulfide and oxygen react with each other, and free sulfide poisons the heme. Riftia's extracellular hemoglobins solve the problem by binding sulfide at a site that is not the heme. Work published in the mid-2000s identified zinc ions on the protein as part of that binding site, so oxygen occupies the iron while sulfide rides safely on the protein, held away from the place where it would do harm. The molecule is a two-cargo carrier with a built-in protective device.

Why should a program on bovine hemoglobin care? Because the problem the tubeworm solved is the general problem of a cell-free carrier: a reactive iron center exposed in plasma, where it meets things it would never meet inside a red cell. The vent worm's solution was to place the hazard on a different part of the protein and to make the protein big and stable. That is an engineering hint, not a treatment. Nobody is going to infuse tubeworm hemoglobin into a patient because it handles sulfide.

What the giants have in common

Set the earthworm, the lugworm, and the tubeworm side by side and a design pattern emerges that has little to do with any one species.

They are large. A few million daltons keeps the molecule in the circulation, out of the kidney, and out of tissue where it could quench nitric oxide.

They are assembled from many chains. Redundancy makes the particle robust: the loss of a single chain does not destroy it, and the many contact points hold the complex together without artificial chemical cross-links.

They are protected by their own scaffold. The linker chains and the metal ions are not decoration. They tune stability, and in the tubeworm they carry a second, dangerous cargo.

They are cooperative. Even at that size they release oxygen on a curve that responds to tissue demand.

Compare this with polymerized bovine hemoglobin. A BHOC is a mixture of chemically linked polymers, with a spread of sizes. Some fraction is small enough to leak, and that fraction is where the historical concern about blood pressure and kidney injury comes from. Manufacturers have pushed to remove low-molecular-weight species for exactly that reason. The natural giants show what happens when the size distribution is not a mixture but a single, defined particle.

The hidden trade

It is tempting to end here with a triumphant claim that worms have solved the problem and the industry should switch. That would be wrong, and the reasons deserve to be spelled out.

The first is immunity. A protein from a marine annelid is more foreign to a human immune system than a protein from a cow. Bovine hemoglobin is still foreign, but cattle proteins have a long history of clinical exposure, and the purified molecule is a globin with a recognizable human-like fold. A single dose of an invertebrate particle may be tolerated. The picture with repeated doses, in patients who need support over days, is far less certain. Antibodies could shorten the half-life or trigger reactions.

The second is supply and cost. Cattle are an established, regulated, large-scale source. Lugworms are harvested from tidal flats. Earthworms are farmed, but the protein has to be extracted from thousands of animals. Any of these can be scaled, but nobody has yet done so at the quantity that a national emergency program would require. The company behind the lugworm product has treated the supply as a proprietary manufacturing process, which is reasonable, and also means outsiders cannot audit it.

The third is oxygen affinity. A carrier that binds oxygen too tightly will not give it up where it is needed, and a carrier that binds too loosely will unload before it reaches the tissue. The natural giants evolved for the temperature, pressure, and oxygen environment of their owners. A worm that lives in cold mud, or on a vent chimney, works at a different oxygen tension from a person in shock. The equilibrium of the protein might need to be tuned, and the assembly is so complex that tuning it is difficult.

The fourth is evidence. Almost everything said about these molecules comes from a handful of laboratories and one developing company. That is normal for an early field. It also means that the claims have not been tested by the adversarial process that eventually broke earlier products.

There is a fifth issue that gets less attention: regulation. A regulator asked to approve a bovine protein has decades of precedent for cattle-derived biologics, including tests for prion

contamination and viral clearance. A regulator asked to approve a worm protein has to write those expectations from scratch. That takes years, and it favors the developers who have already begun. It does not make the science wrong. It does mean that, for a patient who needs a product within this decade, the well-worn path is a real advantage.

Where BHOC fits in this picture

I do not read the natural giants as a threat to the BHOC approach. I read them as a design specification written by evolution.

A bovine hemoglobin product can borrow from the worms in three ways. It can push toward larger, more uniform polymer sizes, which is what several developers are already trying to do. It can add stability through protein-based scaffolds rather than harsh chemical cross-links. And it can pair the carrier with protective elements that buffer its own chemistry, in the spirit of the sulfide-shielding tubeworm.

There is also a second-order use. Giant hemoglobins make useful controls in the laboratory. If a bovine product causes vasoconstriction in an animal model and a giant erythrocrurin does not at a comparable dose, the difference tells you which property of the bovine product matters. That is scientifically worth more than a claim in a brochure.

What to ask when someone tells you nature has solved this

A patient community will hear enthusiastic descriptions of worm blood. A fair listener can ask five plain questions.

How large is the particle, and is it a single defined species or a mixture?

Where did the material come from, and can supply scale without harming an ecosystem?

What happens on the second dose?

How tightly does it hold oxygen, and at what temperature and oxygen tension was that measured?

Who besides the developer has tested it?

None of those questions is hostile. Answering them is what turns a beautiful fact about a worm into a medicine.

The earthworm sitting in a garden carries in its blood a molecule far larger and more elegant than anything a laboratory has made. It has never needed to be a red cell. That is worth knowing, and worth borrowing from, with caution.

Further reading

Primary sources include published biophysical work on *Lumbricus terrestris* erythrocrurin, Hemarina's public descriptions and clinical registrations for its lugworm product, and the mid-2000s structural and biochemical work on sulfide and zinc binding in *Riftia pachyptila* hemoglobins. Figures such as particle mass and chain counts are approximate and should be checked against those sources.